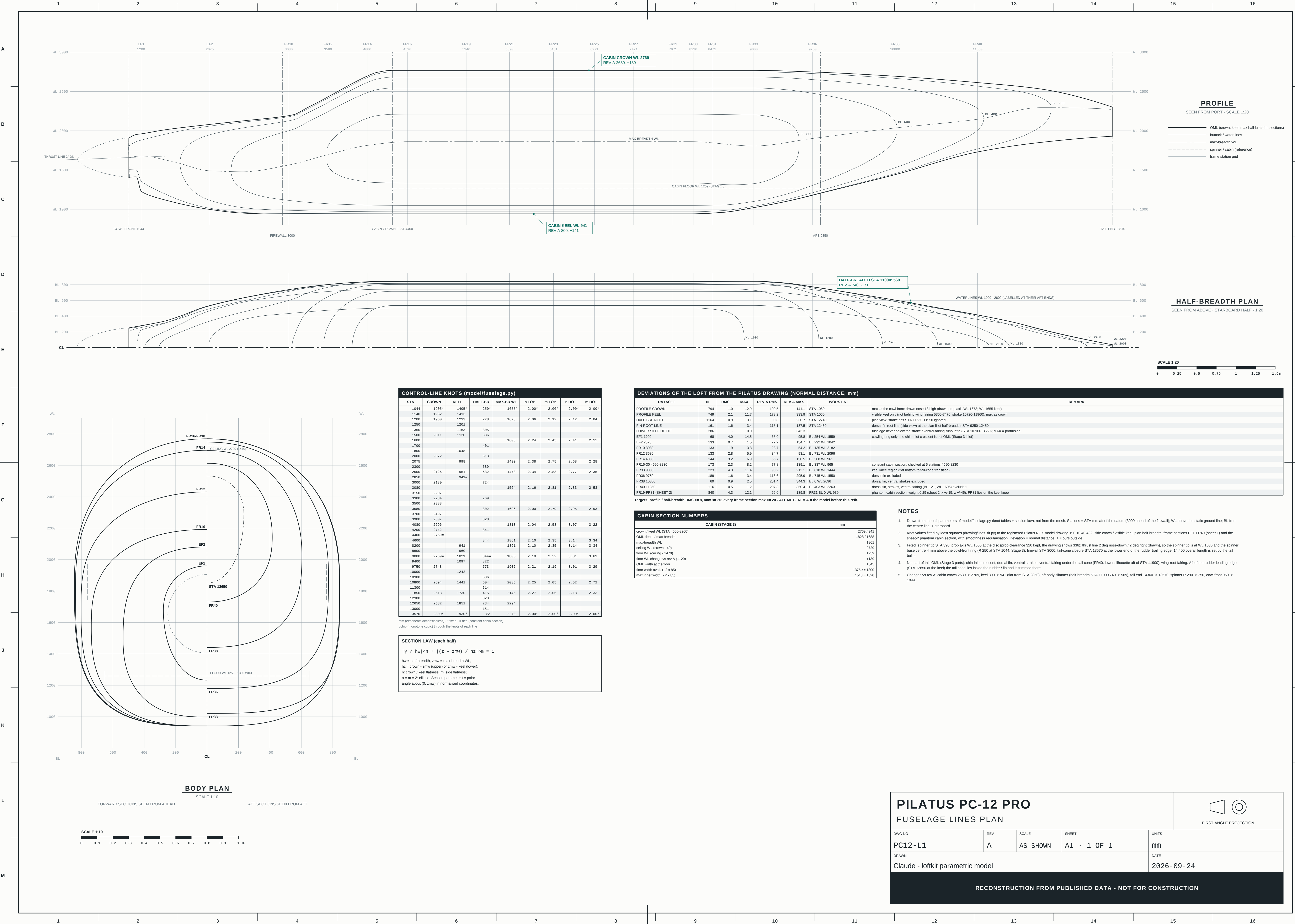




CONTROL-LINE KNOTS (model/fuselage.py)									
STA	CROWN	KEEL	HALF-BR	MAX-BR WL	n TOP	m TOP	n BOT	m BOT	
1844	1985*	1405*	258*	1655*	2.06*	2.06*	2.06*	2.06*	
1148	1952	1413							
1299	1990	1223	278	1678	2.06	2.12	2.12	2.04	
1259		1261							
1359		1163	305						
1509	2911	1129	336						
1709			461	1608	2.24	2.45	2.41	2.15	
1809		1848							
2099	2972		513						
2399		998		1490	2.38	2.75	2.68	2.28	
2509	2126	951	589						
2659		941=	632	1478	2.34	2.83	2.77	2.35	
3099	2180		724						
3899				1564	2.16	2.81	2.83	2.53	
3159	2207								
3399	2284		769						
3599	2385								
3589			802	1696	2.08	2.79	2.95	2.93	
3709	2497								
3999	2697		828						
4999	2695			1813	2.04	2.58	3.07	3.22	
4209	2742		841						
4409	2769=								
4599		941=	844=	1961=	2.18=	2.35=	3.14=	3.34=	
6299				1861=	2.18=	2.35=	3.14=	3.34=	
8699		960							
9099	2769=	1821	844=	1806	2.18	2.52	3.31	3.69	
9499		1997	822						
9759	2748		773	1902	2.21	2.19	3.01	3.29	
10999		1242							
10309			686						
10699	2694	1441	604	2035	2.25	2.05	2.52	2.72	
11399			514						
11859	2613	1730	415	2146	2.27	2.06	2.18	2.33	
12309			323						
12659	2532	1851	234	2294					
13999			151						
13579	2360*	1930*	35*	2279	2.08*	2.08*	2.08*	2.08*	

mm (exponents dimensionless) * fixed - = tied (constant cabin section)
pchip (monotone cubic) through the knots of each line

SECTION LAW (each half)

$|y| / |hw|^n + |(z - zmw) / hz|^m = 1$

hw = half-breadth, zmw = max-breadth WL
hz = crown - zmw (upper) or zmw - keel (lower);
n: crown / keel flatness, m: side flatness;
n = m = 2: ellipse. Section parameter l = polar angle about (0, zmw) in normalised coordinates.

DEVIATIONS OF THE LOFT FROM THE PILATUS DRAWING (NORMAL DISTANCE, mm)								REMARK
DATASET	N	RMS	MAX	REV A RMS	REV A MAX	STA	WORST AT	
PROFILE CROWN	794	1.0	12.9	109.5	141.1	STA 1060		max at the cowl front: drawn nose 18 high (drawn prop axis WL 1673; WL 1655 kept)
PROFILE KEEL	749	2.1	11.7	178.2	333.9	STA 1060		visible keel only (not behind wing fairing 5300-7470; strake 10720-11960); max as crown
HALF-BREADTH	1164	0.9	3.1	90.8	230.7	STA 12740		plan view; strake tips STA 11650-11950 ignored
FIN-ROOT LINE	161	1.6	3.4	118.1	137.5	STA 12450		dorsal-fin root line (side view) at the plan fillet half-breadth; STA 9250-12450
LOWER SILHOUETTE	286	-	0.0	-	343.3			fuselage never below the strake / ventral-fairing silhouette (STA 10700-13560); MAX = protrusion
EF1 1200	68	4.0	14.5	68.0	95.8	BL 254 WL 1559		cowl ring only; the chin-inlet crescent is not OML (Stage 3 inlet)
EF2 2075	133	0.7	1.5	72.2	134.7	BL 292 WL 1042		
FR10 3080	115	1.9	3.8	28.7	54.2	BL 135 WL 2162		
FR12 3580	133	2.8	5.9	34.7	93.1	BL 731 WL 2096		
FR14 4080	144	3.2	6.9	56.7	130.5	BL 308 WL 961		
FR16-30 4590-8230	173	2.3	8.2	77.8	139.1	BL 337 WL 965		constant cabin section, checked at 5 stations 4590-8230
FR33 9000	225	4.3	11.4	90.2	212.3	BL 618 WL 1444		keel knee region (flat bottom to tail-cone transition)
FR38 9750	189	1.6	3.4	116.6	295.9	BL 745 WL 1550		dorsal fin excluded
FR38 10800	69	0.9	2.5	201.4	344.3	BL 0 WL 2696		dorsal fin, ventral strakes excluded
FR40 11850	116	0.5	1.2	207.3	350.4	BL 403 WL 2263		dorsal fin, strakes, ventral fairing (BL 121; WL 1606) excluded
FR19-FR31 (SHEET 2)	840	4.3	12.1	66.0	139.8	FR31; BL 0 WL 939		phantom cabin section, weight 0.25 (sheet 2 - x +/-15, z +/-45); FR31 lies on the keel knee

Targets: profile / half-breadth RMS <= 8, max <= 20; every frame section max <= 20 - ALL MET. REV A = the model before this refit.

CABIN SECTION NUMBERS		
CABIN (STAGE 3)		mm
crown / keel WL (STA 4600-8200)		2769 / 941
OML depth / max breadth		1828 / 1688
max-breadth WL		1861
ceiling WL (crown - 40)		2729
floor WL (ceiling - 1470)		1259
floor WL change vs rev A (1120)		+139
OML width at the floor		1545
floor width avail. (- 2 x 85)		1175 >= 1300
max inner width (- 2 x 85)		1518 - 1520

NOTES

- Drawn from the loft parameters of model/fuselage.py (knot tables + section law), not from the mesh. Stations = STA mm aft of the datum (3000 ahead of the firewall); WL above the static ground line; BL from the centre line, + starboard.
- Knot values fitted by least squares (drawing/lines, fit.py) to the registered Pilatus NGX model drawing 190.10.40.432: side crown / visible keel, plan half-breadth, frame sections EF1-FR40 (sheet 1) and the sheet-2 phantom cabin section, with smoothness regularisation. Deviation = normal distance, + = ours outside.
- Fixed: spinner tip STA 390, prop axis WL 1655 at the disc (prop clearance 320 kept, the drawing shows 336); thrust line 2 deg nose-down / 2 deg right (drawn), so the spinner tip is at WL 1636 and the spinner base centre 4 mm above the cowl-front ring (R 250 at STA 1044; Stage 3); firewall STA 3000, tail-cone closure STA 13570 at the lower end of the rudder trailing edge; 14,400 overall length is set by the tail bullet.
- Not part of this OML (Stage 3 parts): chin-inlet crescent, dorsal fin, ventral strakes, ventral fairing under the tail cone (FR40, lower silhouette aft of STA 11900), wing-root fairing. Aft of the rudder leading edge (STA 12650 at the keel) the tail cone lies inside the rudder / fin and is trimmed there.
- Changes vs rev A: cabin crown 2630 -> 2769, keel 800 -> 941 (flat from STA 2850), aft body slimmer (half-breadth STA 11000 740 -> 569), tail end 14360 -> 13570, spinner R 290 -> 250, cowl front 950 -> 1044.

PILATUS PC-12 PRO

FUSELAGE LINES PLAN

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L1	A	AS SHOWN	A1 · 1 OF 1	mm
DRAWN	DATE			
Claude - loftkit parametric model	2026-09-24			

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION